

**ASSIGNMENT ON TESTING THE NETWORK CONNECTIVITY OF
NETCORE SOLUTIONS USING A CISCO PACKET TRACER**

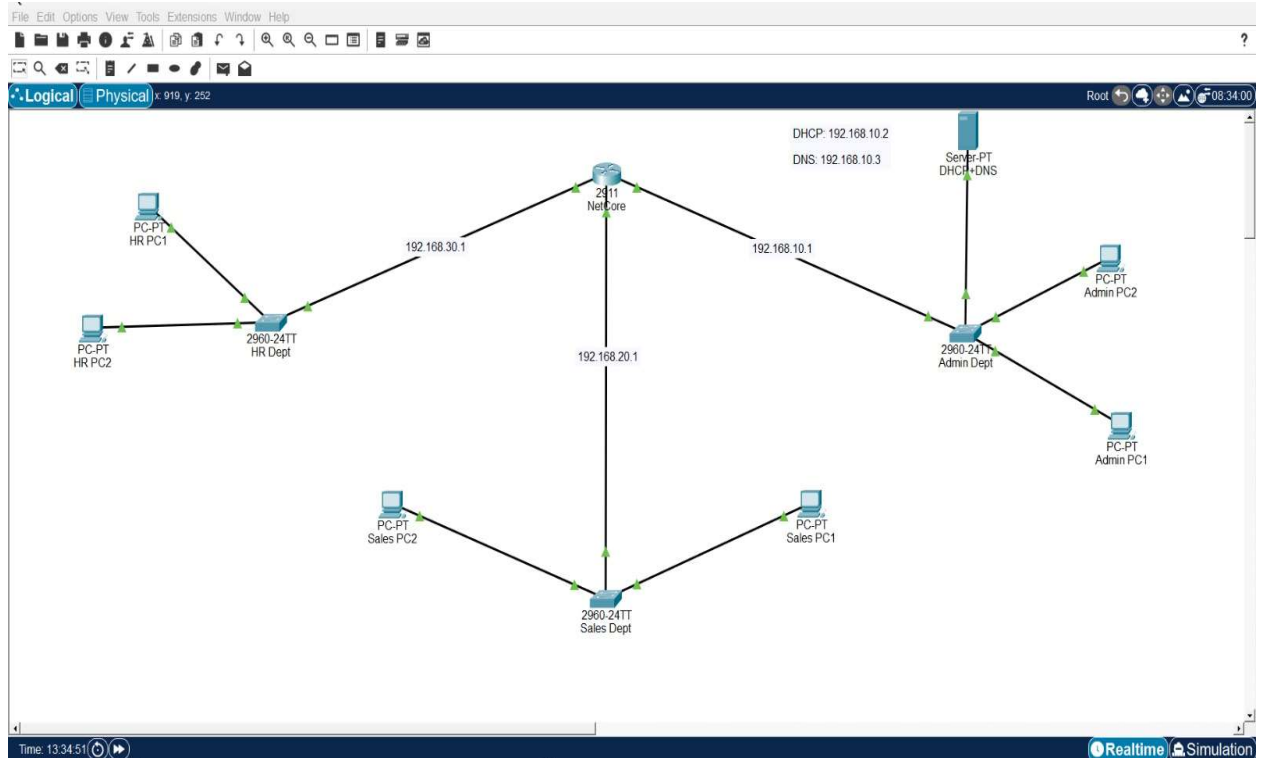
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- 1) The image below shows a screenshot of a “Completed Network Topology” for Netcore Solutions, showing the Admin Dept. Interface, the Sales Dept. Interface and the HR Dept. Interface. The green arrow signs and the topography clearly show that the devices are connected, that all IP addresses have been assigned successfully, and that they are running fine. The router has also been configured so that the server can automatically assign an IP address to any PC in the Sales and HR departments. This shows 1 Router (2911), 3 switches (1 per department) and 6pcs (2 per department), and 1 server as indicated in the project workflow.



2) A- Admin Pc 2 IP Address Configuration

The image shows the IP address assigned to Admin PC1 automatically as the DHCP server is automatically related to it.

The screenshot shows a window titled "Admin PC2" with tabs for Physical, Config, Desktop, Programming, and Attributes. The "Desktop" tab is selected, and the "IP Configuration" window is open. The "Interface" dropdown is set to "FastEthernet0".

IP Configuration

Interface: FastEthernet0

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address	192.168.10.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	192.168.10.3

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address	
Link Local Address	FE80::202:17FF:FE50:C037
Default Gateway	
DNS Server	

802.1X

☐ Use 802.1X Security

Authentication	MD5
Username	
Password	

☐ Top

B – Sales PC2 IP Address Configuration

The image shows the IP Address of Sales PC1 assigned after the DHCP interface has been configured with the use of “ip helper” to connect the DHCP Server to the sales departments.

The screenshot displays the configuration window for Sales PC2, specifically the Desktop tab. The IP Configuration section is active, showing settings for the FastEthernet0 interface. The DHCP option is selected, and the IPv4 Address is 192.168.20.11. The Subnet Mask is 255.255.255.0, the Default Gateway is 192.168.20.1, and the DNS Server is 192.168.10.3. The IPv6 Configuration section is also visible, with the Static option selected and the Link Local Address set to FE80::210:11FF:FECE:430A. The 802.1X section shows the Use 802.1X Security checkbox unchecked, with Authentication set to MD5. The Username and Password fields are empty. A Top button is located at the bottom left.

Sales PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.20.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.20.1

DNS Server 192.168.10.3

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::210:11FF:FECE:430A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

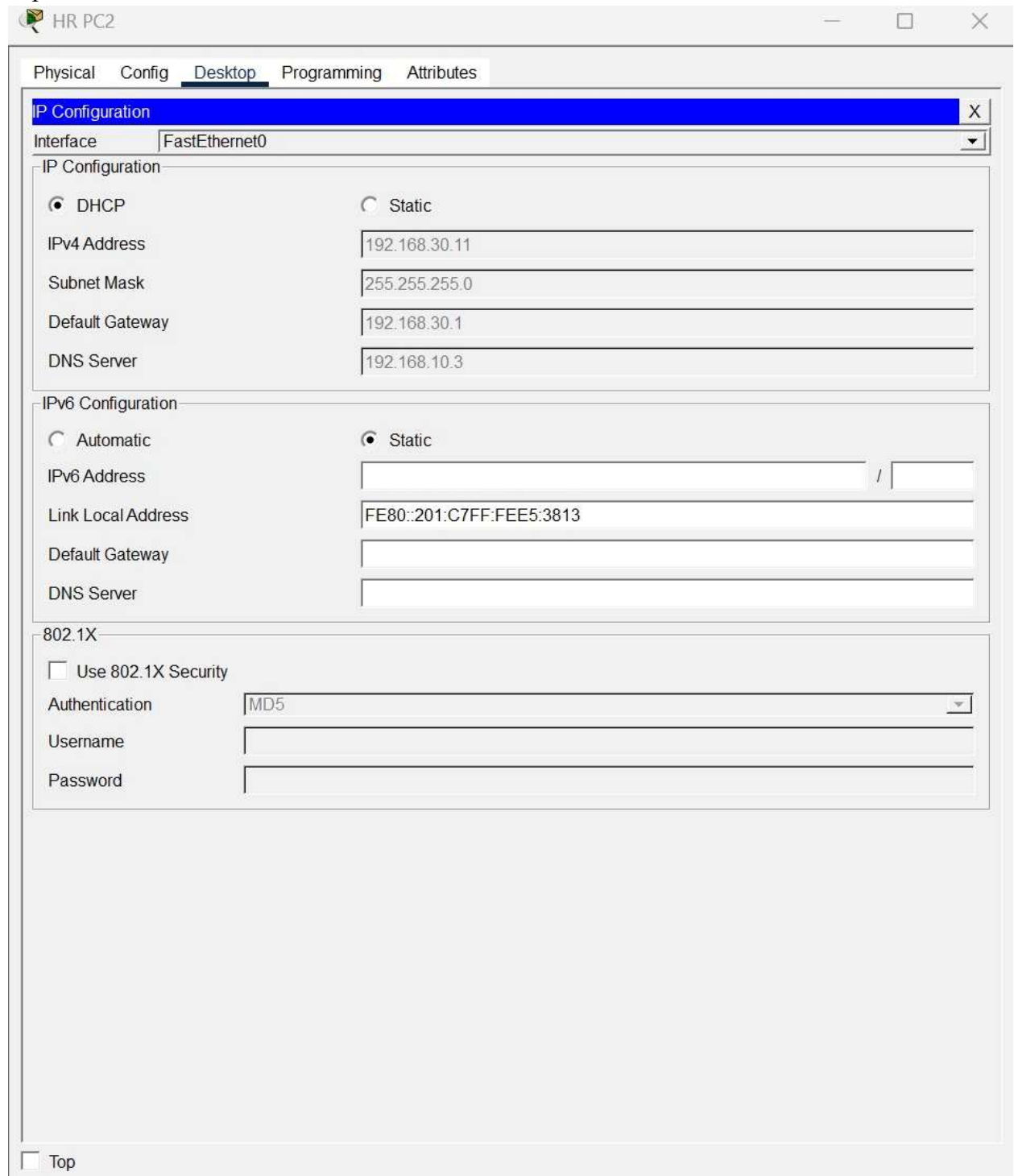
Username

Password

☐ Top

C – HR PC2 IP Address Configuration

The image shows the IP Address of HR PC1 assigned after the DHCP interface has been configured with the use of “ip helper” to connect the DHCP Server to the sales departments.



The screenshot displays the configuration window for HR PC2, specifically the Desktop tab. The IP Configuration section is active, showing settings for the FastEthernet0 interface. The DHCP option is selected, and the IPv4 Address is set to 192.168.30.11. The Subnet Mask is 255.255.255.0, the Default Gateway is 192.168.30.1, and the DNS Server is 192.168.10.3. The IPv6 Configuration section shows the Static option selected, with a Link Local Address of FE80::201:C7FF:FEE5:3813. The 802.1X section is also visible, with the Use 802.1X Security checkbox unchecked and the Authentication dropdown set to MD5.

HR PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 192.168.30.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.30.1

DNS Server 192.168.10.3

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C7FF:FEE5:3813

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

3) A successful Ping from the Sales PC to the Admin PC.

The image shows a successful connection/interaction/communication from the sales department to the admin departments. This means the sales department is able to communicate successfully with the admin departments, which is in line with the project's objective to have successful communication from sales to admin.

The first ping indicates 1 packet was lost in the process of communication, being the first time and was fully successful with a loss of a packet in the second communication attempt.

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.11: bytes=32 time=11ms TTL=127
Reply from 192.168.10.11: bytes=32 time=2ms TTL=127
Reply from 192.168.10.11: bytes=32 time=13ms TTL=127

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 13ms, Average = 8ms

C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time=11ms TTL=127
Reply from 192.168.10.11: bytes=32 time<1ms TTL=127
Reply from 192.168.10.11: bytes=32 time=17ms TTL=127
Reply from 192.168.10.11: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 7ms

C:\>
```

4) A failed ping from the HR PC to an Admin PC

The image shows an unsuccessful connection/communication from the HR department to the admin department. This means an ACL command has been configured on the router to restrict the communication from HR departments to the admin departments. No communication or interaction can be successfully processed from the HR departments to the admin departments, which is in line with the project's objective to have failed communication from HR to Admin.

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

5) Screenshot of the ACL Command

The image shows an ACL command that shows the configuration assigned to deny communication from the HR department to the Admin Department, and also permits communication from the HR department to the Sales department, while permitting any other external IP addresses.

```
netcore>enable
netcore#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
netcore(config)#ip access-list extended HR-ACL
netcore(config-ext-nacl)#deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
netcore(config-ext-nacl)#permit ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
netcore(config-ext-nacl)#permit ip any any
netcore(config-ext-nacl)#exit
netcore(config)#interface GigabitEthernet0/2
netcore(config-if)#ip access-group HR-ACL in
netcore(config-if)#exit
netcore(config)#end
netcore#
%SYS-5-CONFIG_I: Configured from console by console

netcore#show access-lists
Extended IP access list HR-ACL
 10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 20 permit ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 30 permit ip any any
netcore#
```